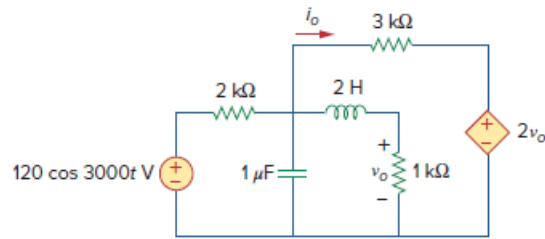


Problem

Use *PSpice* to obtain v_o and i_o in the circuit of Fig. 10.37.

Figure 10.37:



Step-by-step solution

Step 1 of 7

Step (1)

The frequency ' f ' is obtained from ω as

$$f = \frac{\omega}{2\pi}$$

$$f = \frac{3000}{2 \times 3.1416}$$

$$f = 477.464 \text{ Hz}$$

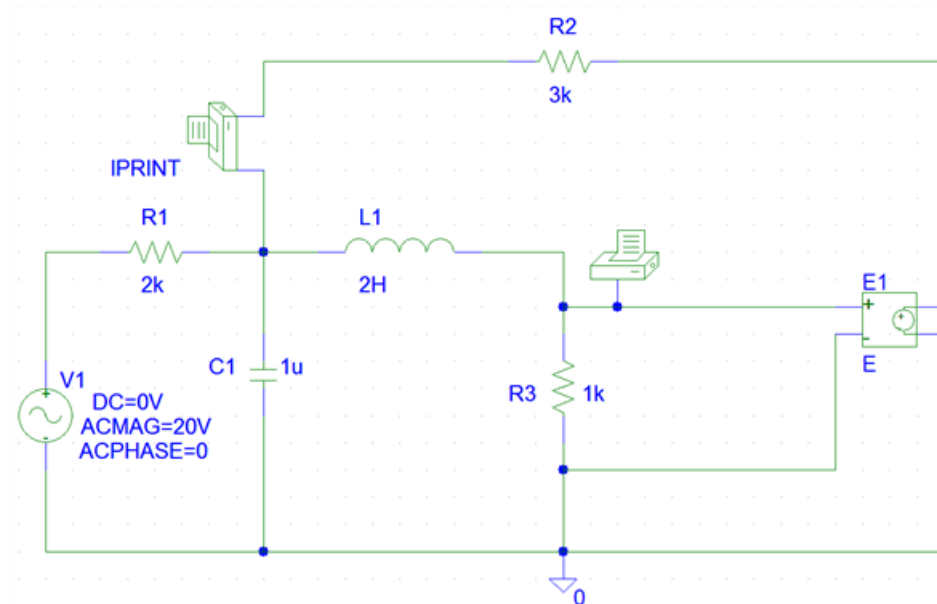
Step 2 of 7

Step (2)

Open a new Schematic and connect all the components as shown in below circuit. Change all components attributes by double clicking on them. By double click as AC voltage source and set DC = 0V, AC MAG = 20 V and AC PHASE = 0

.Notice how the voltage controlled voltage source is connected

Step 3 of 7



Step (3)

We only want the magnitude and phase of v_o and i_o , we set the attributes of IPRINT and VPRINT each to AC = yes, MAG = yes, PHASE = yes

Step 4 of 7

Step (4)

Goto 'Analysis' menu open setup / AC sweep and enter total pts = 1, start freq = 477.464 and Final Freq = 477.464

Step 5 of 7

Step (5)

After saving the schematic, we simulate it by selecting analysis / simulate. The output file includes the source frequency in addition to the attributes checked for the pseudo components IPRINT and VPRINT as shown below

Step 6 of 7

```
FREQ IM(V_PRINT1) IP(V_PRINT1)
4.775E+02 1.088E-03 -5.512E+01
FREQ VM($N_0003) VP($N_0003)
4.775E+02 5.366E-01 -1.546E+02
```

Step 7 of 7

From this output file, we obtain

$$v_0 = 536.6 \angle -154.6^\circ \text{ mV}$$

$$i_0 = 1.088 \angle -55.12^\circ \text{ mA}$$

Which are the phasors for

$$v_0(t) = 536.6 \cos(3000t - 154.6^\circ) \text{ mV}$$

And

$$i_0(t) = 1.088 \cos(3000t - 55.12^\circ) \text{ mA}$$